MLR780 Air Conditioning and Refrigeration 2021

Assignment 8
Energy Storage

Instructions

- This individual assignment is to be submitted online before midnight on **20 June 2021**.
- Repetitive calculation will be needed. Use a programming platform or a spread sheet of your choice. Use CoolProp for thermodynamic properties.
- Prepare a technical report that supplies all relevant equations, methods and input values, and results. Include everything that is to be graded in your report. Name it: "MLR780-2021-Assignment8-SURNAME.pdf"
- Each submission (PDF) **must** be accompanied by an official cover page obtainable from Click-UP, **as well as** your supporting documentation to prove you performed the analysis.
- The supporting documentation will not be graded, but will be checked: It must be a soft version of the script (a *.txt text file) or spread sheet with the following naming convention: "MLR780-2021-Assignment8-SURNAME.* " (The "*" represent the file extension, which will depend on the program you used – it should not be "pdf").
- **Only the PDF report will be graded.**
- Keep a time sheet documenting the time it took you to solve the questions and indicate this on the first page of your submission. Time spent on the lecture content / example / programming revision should not be included.

Introduction

Consider a cooling load profile, as supplied in Table 1. These are the average cooling coil requirement.

Table 1 Cooling Loads

Hour	Cooling Load	Hour	Cooling Load
	W		**W**
0	1833	12	9383
1	1584	13	9963
2	1373	14	11 100
3	1029	15	12 200
4	908	16	14 791
5	404	17	15 102
6	480	18	12 258
7	1776	19	10 304
8	3124	20	9840
9	6235	21	8457
10	7653	22	5517
11	8820	23	2600

Due to power interruption schedules (load shedding), an ice thermal storage system (external melt on tube) is to be incorporated. During times of power interruption, back-up power is supplied only to the variable speed pumps, fans and the control systems,

but not to the chiller, which for this installation accounts for the largest energy consumption of the air-conditioning system.

Consider the following two schedules when there will not be power to the chiller:

- Schedule A: 08:00 – 10:00 and 20:00 – 22:00
- Schedule B: 02:00 – 04:00, 10:00 – 12:00 and 18:00 – 20:00

For purposes of this assignment, assume that the chiller has no speed control and can either be on (x = 1) or off (x = 0). It can run for all hours except the hours listed above, and only when the thermal storage is not fully charged. It should switch of when the thermal storage is fully charged.

The thermal storage must be able to fully sustain the required cooling for all hours when the chiller is switched off. Ignore heat transfer to the surrounding.

1.1 Perform a first-order thermal storage sizing exercise using only the energy balance principle **to specify the minimum required thermal storage capacity** (E_s) in [J], as well as the required chiller rating ($\dot{Q}_c$) [W]. Take both Schedule A and B into consideration and for each hour calculate the required storage charging rate ($\dot{E}_s^+$) [W], required storage discharge rate ($\dot{E}_s^-$) [W], the cooling energy stored in the ice-system (E) at the end of each hour, as well the level of charge of the thermal storage charge $G \epsilon$ [0;1] ($G = E/E_s$). G must be equal to 1 at least once in a 24-hour cycle. In this question the on/off decision (toggling the value of x) can only be made once, at the start of every hour.

Hint: the method used in the class example should **not** be followed blindly. Some fundamental adjustments are needed to solve this question correctly.

- Tabulate x, $\dot{Q}_c$, $\dot{E}_s^+$, $\dot{E}_s^-$, E, and G for hours 0 to 23
- Show your calculations in full for Schedule A between 17:00 and 19:00.
- Plot E against time.

Now consider the ice storage system. The secondary refrigerant temperature (water) returning from the cooling coil is 15ºC (T_{in}). For this inlet temperature, the discharge effectiveness will depend on the fraction of storage discharged, $F \epsilon$ [0;1] ($F = 1 - G$), as well as the secondary refrigerant mass flow rate, $\dot{m}$. This effectiveness expresses the actual discharge rate as a fraction of the maximum theoretical discharge rate. Two discharge performance curves (at secondary refrigerant mass flow rates of 0.05 kg/s and 0.3 kg/s) are available: see equation 1 and the accompanied coefficients in Table 2. Note that this is valid only for this particular type of storage system from a particular manufacturer, and has been based on experimental data.

$$\varepsilon = C_0 + C_1 F + C_2 F^2 + \cdots + C_5 F^5 \quad (1)$$

Table 2 Discharge effectiveness polynomial coefficients.

	Curve I	Curve II
$\dot{m}$	0.05 kg/s	0.3 kg/s
C_0	1	1
C_1	-0.43	-1.8
C_2	3.85	9.1
C_3	-15.91	-25.4
C_4	23.74	32.0
C_5	-12.25	-14.9

1.2 Based only on latent heat, specify the mass of ice to match the required cooling capacity you derived in question 1.1.

1.3 For the specified mass of ice (question 1.2), calculate the required secondary refrigerant mass flow rate (it is allowed to be any value between 0 kg/s and 0.5 kg/s) that is needed to supply the relevant discharge rate as determined in question 1.1. Perform this discharge analysis in intervals 15 minutes, and update the required mass flow rate for each interval. (Thus, recalculate the flow rate for every 15 minutes). In this question the on/off decision (toggling the value of x) can be made once every 15 minutes.

Due to a lack of information, assume the required discharge rate (linked to rates in Table 1) remains constant within each hour (thus, 4 successive discharge intervals will have the same required cooling rate). Adopt a linear interpolation / extrapolation approach between curve I and II if needed. Ignore heat transfer to the surrounding. Include a first order energy balance analysis to keep track of the ice storage recharge status during the times when the chiller is operational, and use the relevant charged fraction of each discharge period.

- Show your calculations (with equations) in full for Schedule B between 10:00 and 10:30 (two 15-minute intervals) for: $\dot{m}$ (required operating secondary refrigerant mass flow rate), ε (applied effectiveness for that interval), $\dot{E}_s^+$, $\dot{E}_s^-$, F (fraction of storage discharged at the end of that time interval), and total cooling supplied to the cooling coil, $\dot{Q}$.
- Tabulate the time in hours, x, $\dot{m}$, $\dot{E}_s^+$, $\dot{E}_s^-$, F and $\dot{Q}$ for all time intervals between 08:00 and 17:00 for both Schedule A and B.
- Plot your results against time.

1.4 Comment on whether the ice mass calculated in question 1.2 is sufficient. If not sufficient, you are **not** required to repeat the analysis to determine the ideal ice mass, but you are required to make suggestions on how to address the problem.

INDICATE THE TIME YOU SPENT TO COMPLETE THIS ASSIGNMENT BY PUTTING IT ON YOUR COVER PAGE.